

VANSH CHAUDHARY

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EXPERIENCE

AI/ML & DATA ANALYST TRAINEE

- Built and trained deep learning models using CNNs and RNNs for real-world classification tasks
- Worked with Large Language Models (LLMs) and generative AI tools, applying modern NLP techniques
- Applied supervised ML (XGBoost, Random Forest, LSTM) with feature selection, cross-validation, and hyper parameter tuning to predict fuel efficiency and failure rates across 500+ BMW models — achieving 92% ROC-AUC

EDUCATION

B.Tech in Computer Science & Engineering Bahra University, Solan	CGPA 7.4 / 10.0
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SKILLS

Technical

Python AI/ML	Hugging Face	SQL	AWS	Excel
TensorFlow/Keras	OpenCV	Scikit-learn	Power BI	

Soft Skills

Teamwork	Communication	Problem-Solving
Analytical Thinking		

Certifications

Data Science Methodology	AI/ML certification
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Certifications Link

<https://drive.google.com/drive/folders/1pxDuZthWegj4D5DHDP1Uap8mJ5ULGVr3?usp=sharing>

Projects Links

- <https://github.com/VanshChaudhary-11/Speech-model>
- <https://github.com/VanshChaudhary-11/Speech-model/blob/main/speechmodel.py>
- [https://github.com/VanshChaudhary-11/DATA-ANALYST/blob/main/BMW\(adv\)11.html](https://github.com/VanshChaudhary-11/DATA-ANALYST/blob/main/BMW(adv)11.html)

SUMMARY

Computer Science graduate with hands-on experience in machine learning and data analysis. Skilled at cleaning complex datasets, building and tuning predictive models, and translating raw data into clear, meaningful insights

PROJECTS

Speech emotion recognition

- Developed a speech emotion recognition module for a multimodal AI interview coaching platform, extracting MFCC, chroma, mel-spectrogram, RMS, and zero-crossing-rate features from audio using Librosa.
- Trained a Random Forest classifier (tuned with 200 estimators, max depth, and feature subsampling to classify 8 emotion categories from extracted acoustic features.
- Evaluated model performance using an 80/20 train-test split, n classification accuracy as a key metric.

Real-time facial emotion recognition

- Built a real-time facial emotion recognition system using OpenCV and DeepFace, combining Haar Cascade face detection with deep learning-based emotion classification.
- Captured live webcam feed, detected faces per frame, and ran DeepFace's emotion analysis model on each detected face region (ROI).
- Optimized performance by running emotion inference every 5th frame instead of every frame, while caching and displaying the last detected label continuously for smooth UI feedback.

Analytics for BMW Vehicle across various regions.

- Developed predictive ML models on 500 BMW vehicle datasets to forecast fuel efficiency, engine wear, and failure rates, reducing predicted maintenance costs by 15% and achieving 92% accuracy in early failure detection
- Performed full ML pipeline: data acquisition, EDA, preprocessing, model development, and visualization
- Delivered an interactive Jupyter Notebook dashboard for stakeholders highlighting degradation factors and maintenance strategies.